Models of the motor system

Fall 2021

Opher Donchin

Part 1

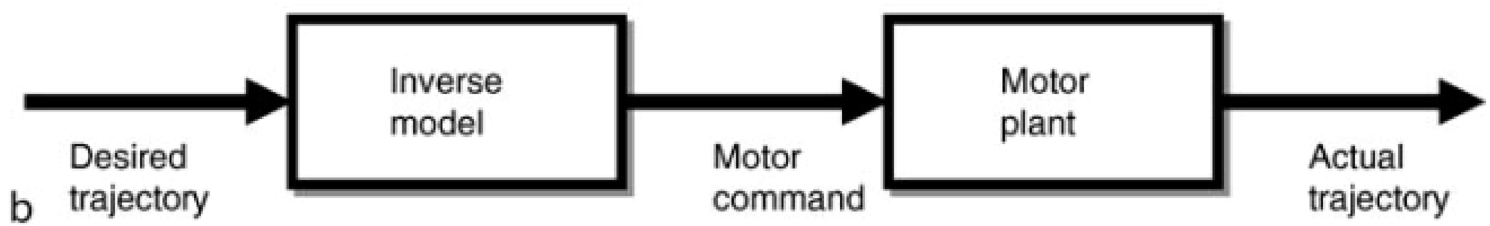
- Internal models

Internal models

- Brain calculations of kinematics and dynamics

Inverse internal model

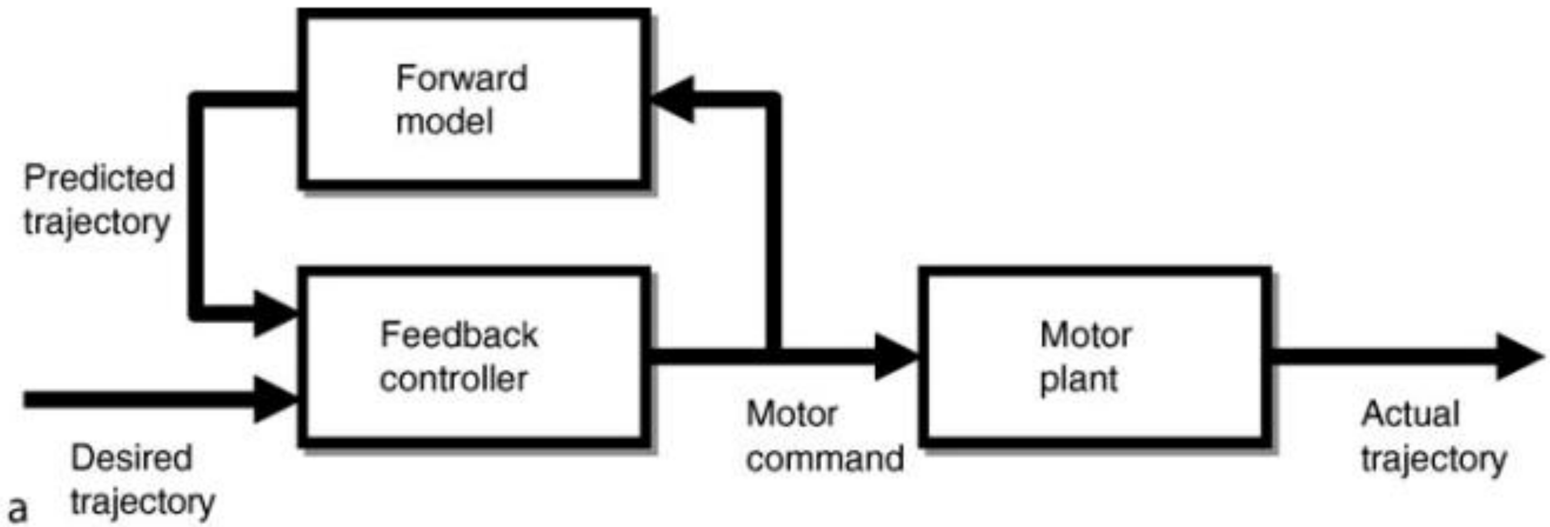
- Invert the forward dynamics and kinematics



Cisek *Encyclopedia of Neuroscience* 2009 doi:10.1007/978-3-540-29678-2_2528

Forward internal model

- Simulate the forward dynamics and kinematics



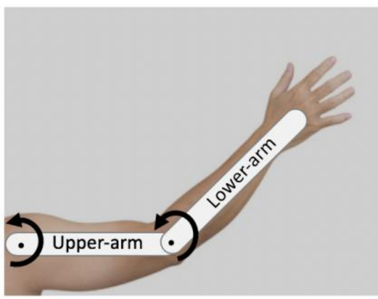
Cisek *Encyclopedia of Neuroscience* 2009 doi:10.1007/978-3-540-29678-2_2528

Part 2

- Interaction torques

Intersegmental dynamics

- Interaction moments
  - Torques at one joint due to motion of another joint



$$\ddot{\theta} = M^{-1}\left(\tau - C\dot{\theta} - G\right)$$

$$M = \begin{pmatrix} I_1 + I_2 + m_1 r_1^2 + m_2\left(l_1^2 + r_2^2 + 2 l_1 r_2 \cos\theta_e\right) & I_2 + m_2\left(r_2^2 + l_1 r_2 \cos\theta_e\right) \\ I_2 + m_2\left(r_2^2 + l_1 r_2 \cos\theta_e\right) & I_2 + m_2 r_2^2 \end{pmatrix}$$

$$C = l_1 m_2 r_2 \sin\theta_e \begin{pmatrix} 0 & -2\dot{\theta}_s - \dot{\theta}_e \\ \dot{\theta}_s & 0 \end{pmatrix}$$

Coriolis effects across joints

- If there are no torques
- And only the shoulder is moving
- And ignore gravity

$$\tau = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \dot{\theta} = \begin{pmatrix} \dot{\theta}_s \\ 0 \end{pmatrix} \qquad C = l_1 m_2 r_2 \sin\theta_e \begin{pmatrix} 0 & -2\dot{\theta}_s - \dot{\theta}_e \\ \dot{\theta}_s & 0 \end{pmatrix}$$

$$G = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\ddot{\theta} = M^{-1}\left(\tau - C\dot{\theta} - G\right) = M^{-1}\left(-C\dot{\theta}\right) = -M^{-1} l_1 m_2 r_2 \sin\theta_e \begin{pmatrix} 0 \\ \dot{\theta}_s^2 \end{pmatrix}$$

The rotation of the shoulder acts like a torque opening the elbow

Coriolis effects across joints

- If there are no torques
- And only the **elbow** is moving
- And ignore gravity

$$\tau = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \dot{\theta} = \begin{pmatrix} 0 \\ \dot{\theta}_e \end{pmatrix} \qquad C = l_1 m_2 r_2 \sin\theta_e \begin{pmatrix} 0 & -2\dot{\theta}_s - \dot{\theta}_e \\ \dot{\theta}_s & 0 \end{pmatrix}$$

$$G = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

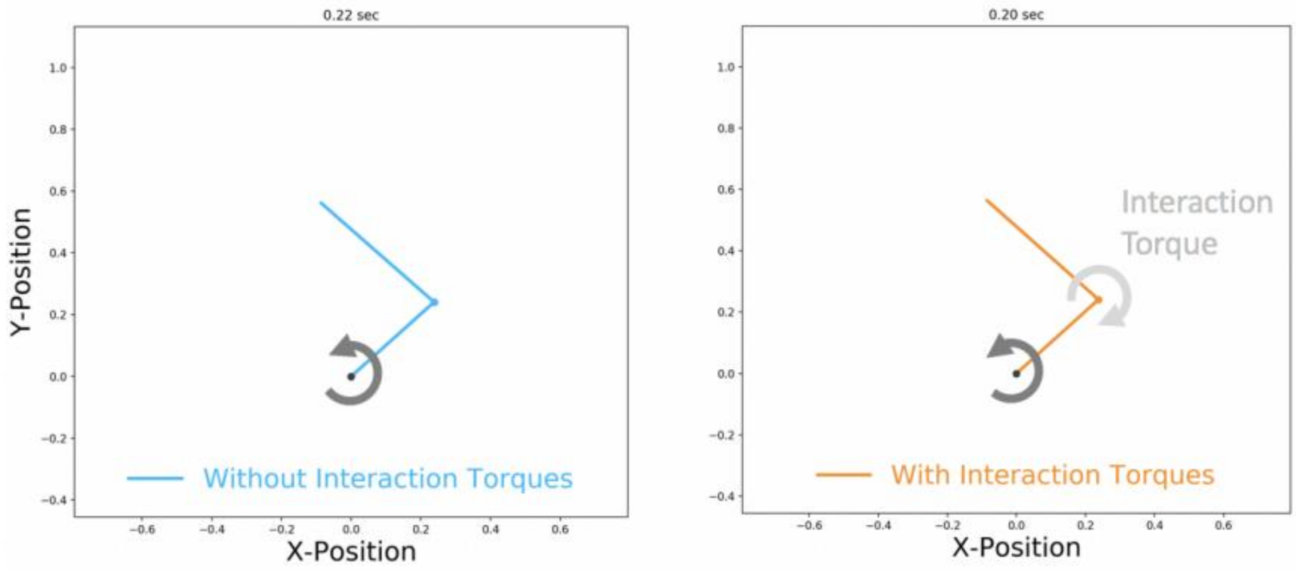
$$\ddot{\theta} = M^{-1}\left(\tau - C\dot{\theta} - G\right) = M^{-1}\left(-C\dot{\theta}\right) = M^{-1} l_1 m_2 r_2 \sin\theta_e \begin{pmatrix} \dot{\theta}_e^2 \\ 0 \end{pmatrix}$$

The rotation of the elbow acts like a torque closing the shoulder

If elbow is passive

- Shoulder flexion would cause elbow extension

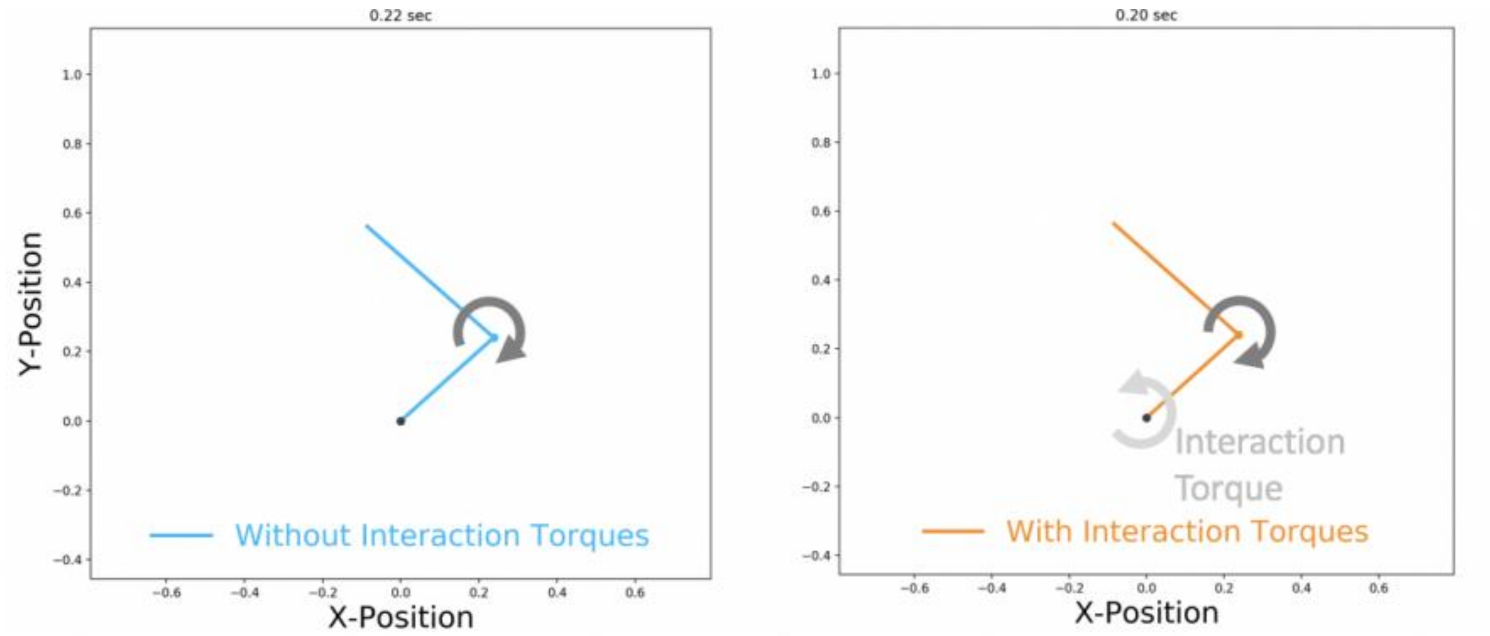
$$Q_1 \rightarrow \dot{\theta}_1, \ddot{\theta}_1 \rightarrow Q_2 \rightarrow \dot{\theta}_2, \ddot{\theta}_2$$



Shoulder rotation leads to elbow rotation

If shoulder is passive

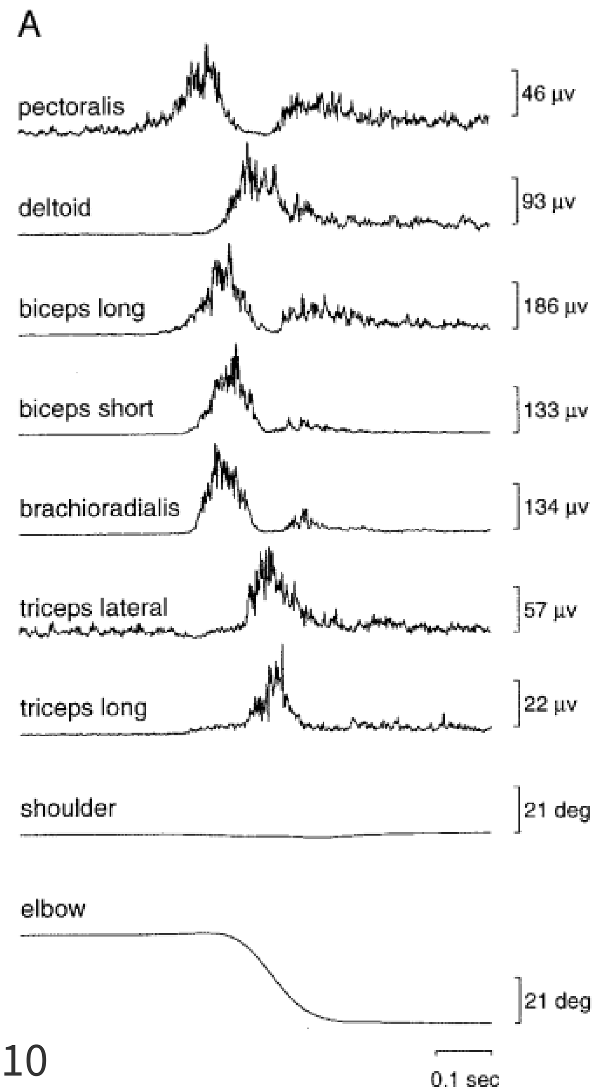
- Elbow flexion causes shoulder extension



Elbow rotation leads to shoulder rotation

Motor system compensates for interaction torques

- Single-joint elbow movement
  - Activation of shoulder muscles
  - No shoulder movement



Gribble *J Neurophys* 1999 doi:10.1152/jn.1999.82.5.2310

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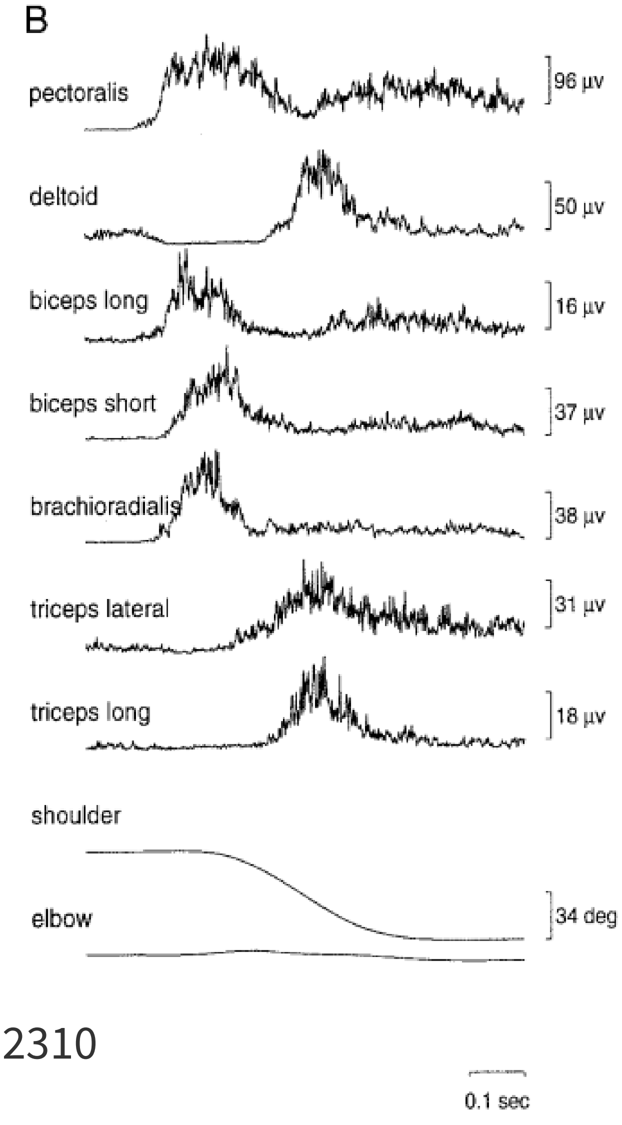
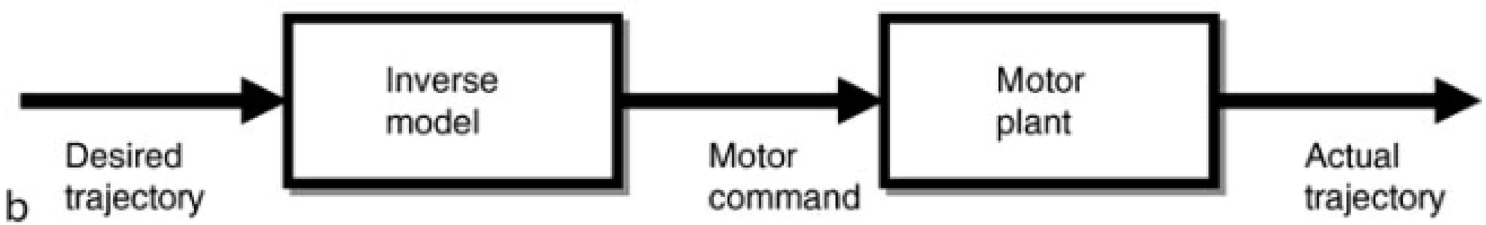


Gribble *J Neurophys* 1999 doi:10.1152/jn.1999.82.5.2310

Evidence of internal model

- Pre-planning for interaction torques
  - Internal inverse model of dynamics

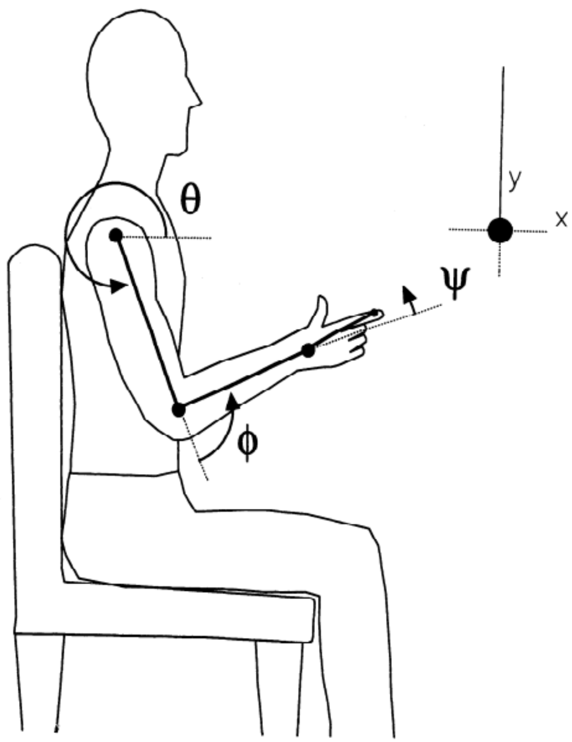


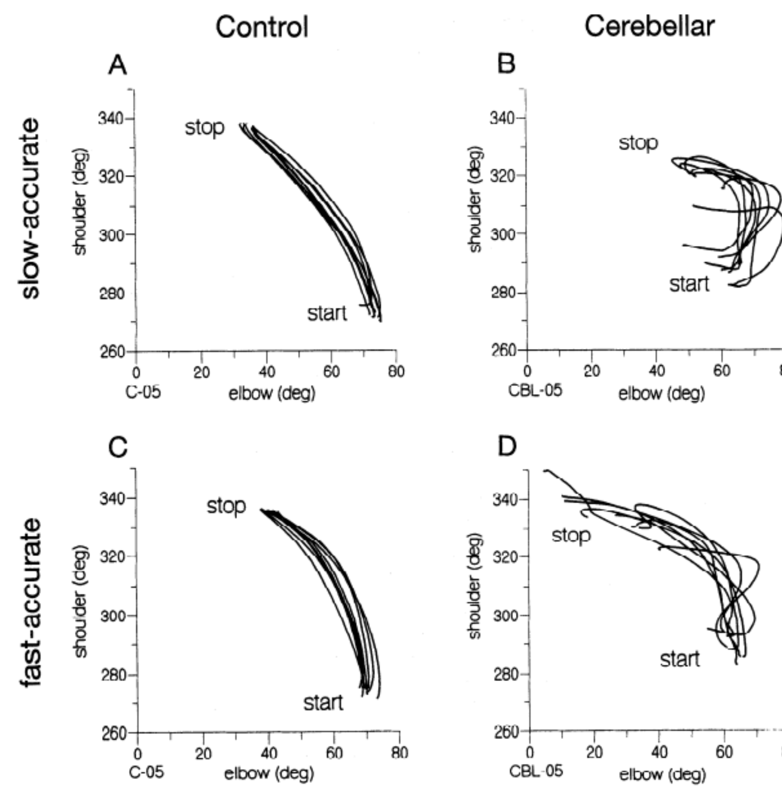
Part 3

- Cerebellar ataxia

Cerebellar disorder

- Ataxia
  - Disjoint movements
  - Lack of cooordination

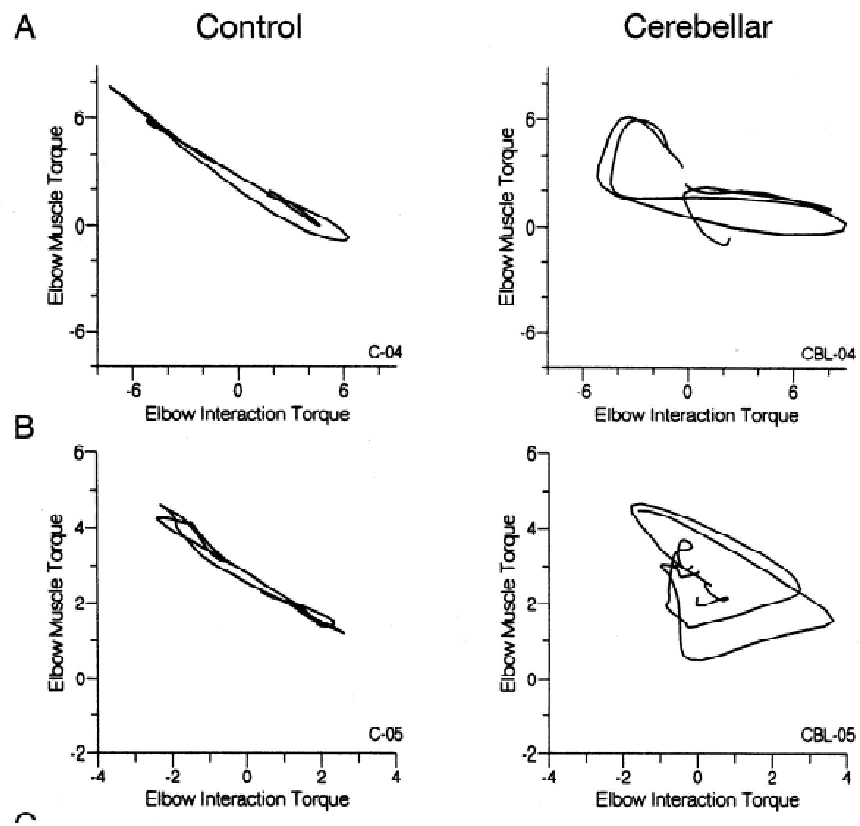




Bastian *J Neurophys* 1996 doi:10.1152/jn.1996.76.1.492

Ataxia reflects failure to compensate for interaction torques

- Decompose forces on elbow
  - Muscle forces
  - Interaction forces
- Cerebellar patients fail to compensate



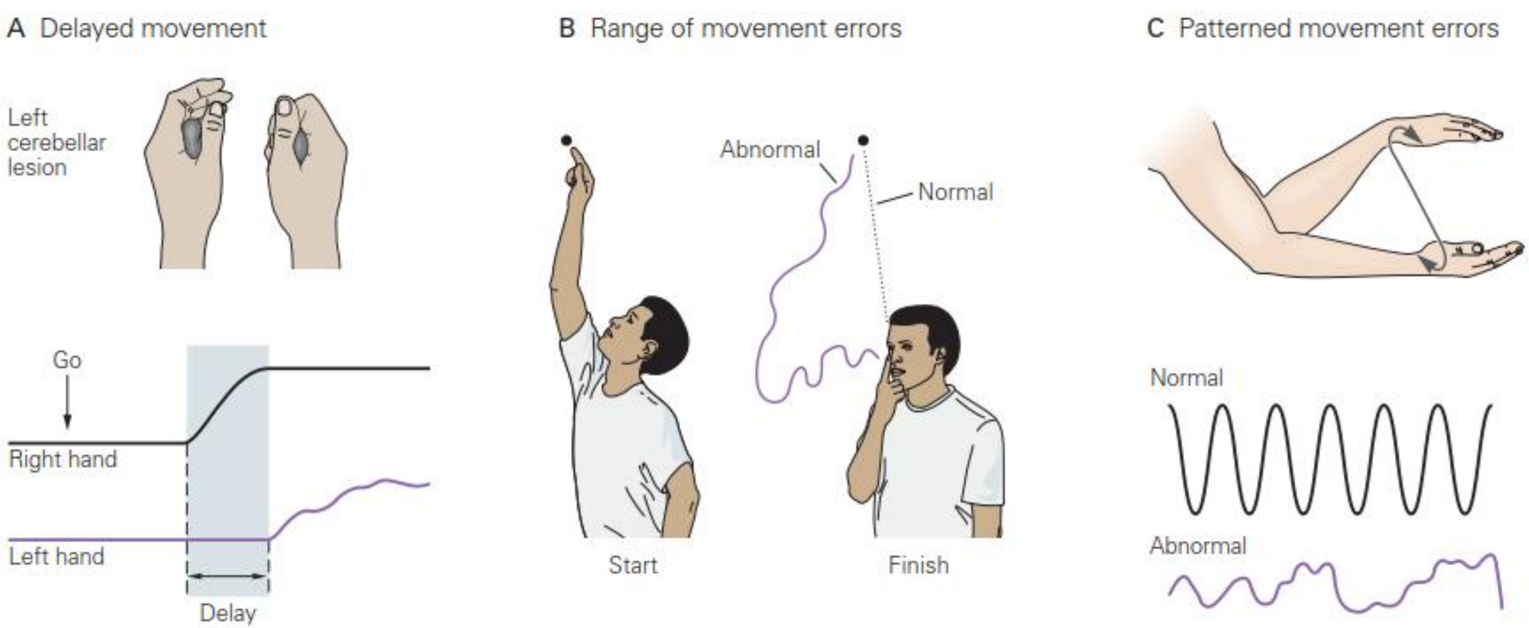
Bastian *J Neurophys* 1996 doi:10.1152/jn.1996.76.1.492

Cerebellar ataxia

- Caused by
  - Stroke
  - Trauma
  - Cellular degeneration (genetic)
  - Excessive alcohol
- Symptoms
  - Intention tremor
  - Dysmetria
  - Slurred speech
  - Balance

Cartoon version

- Textbook figure



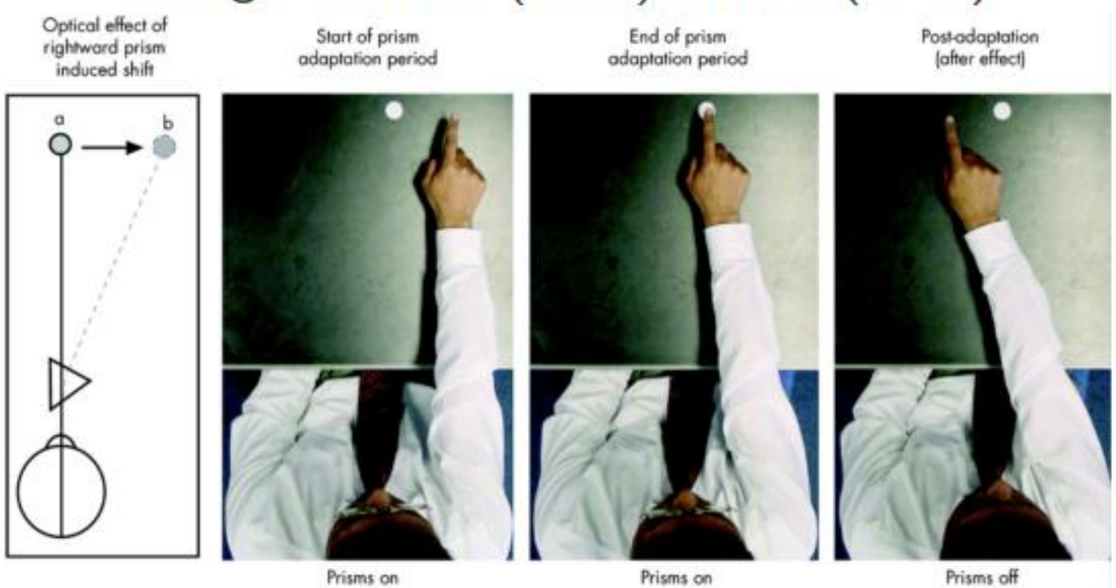
Demonstration of ataxia

Talking with a patient

Prism adaptation

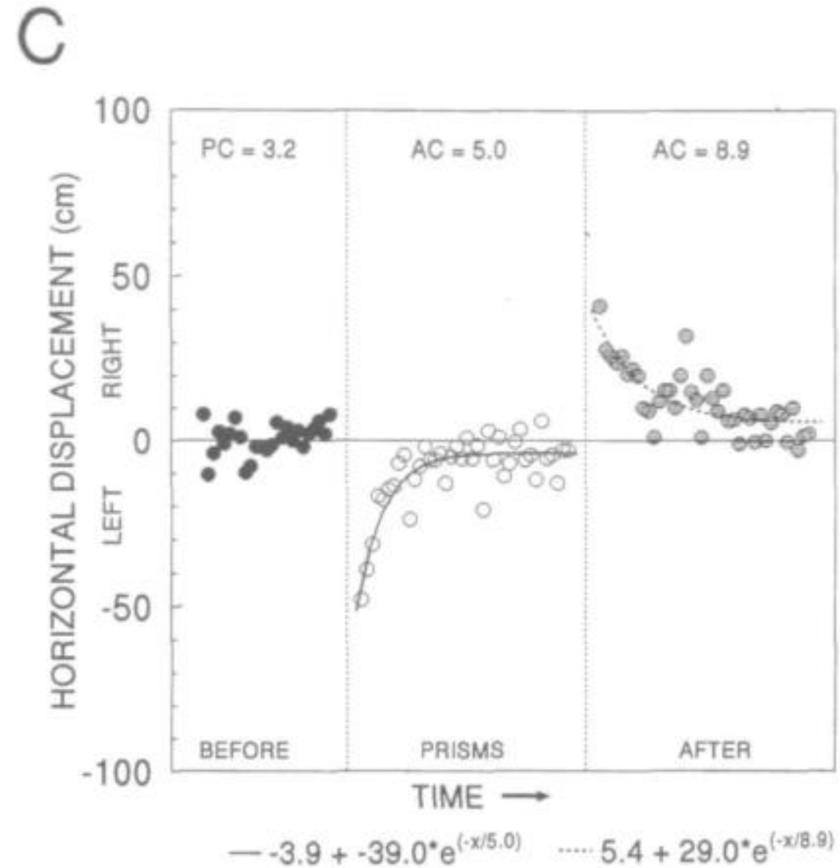
"Visuomotor Rotation"

George Stratton (1896) Kohler (1951)



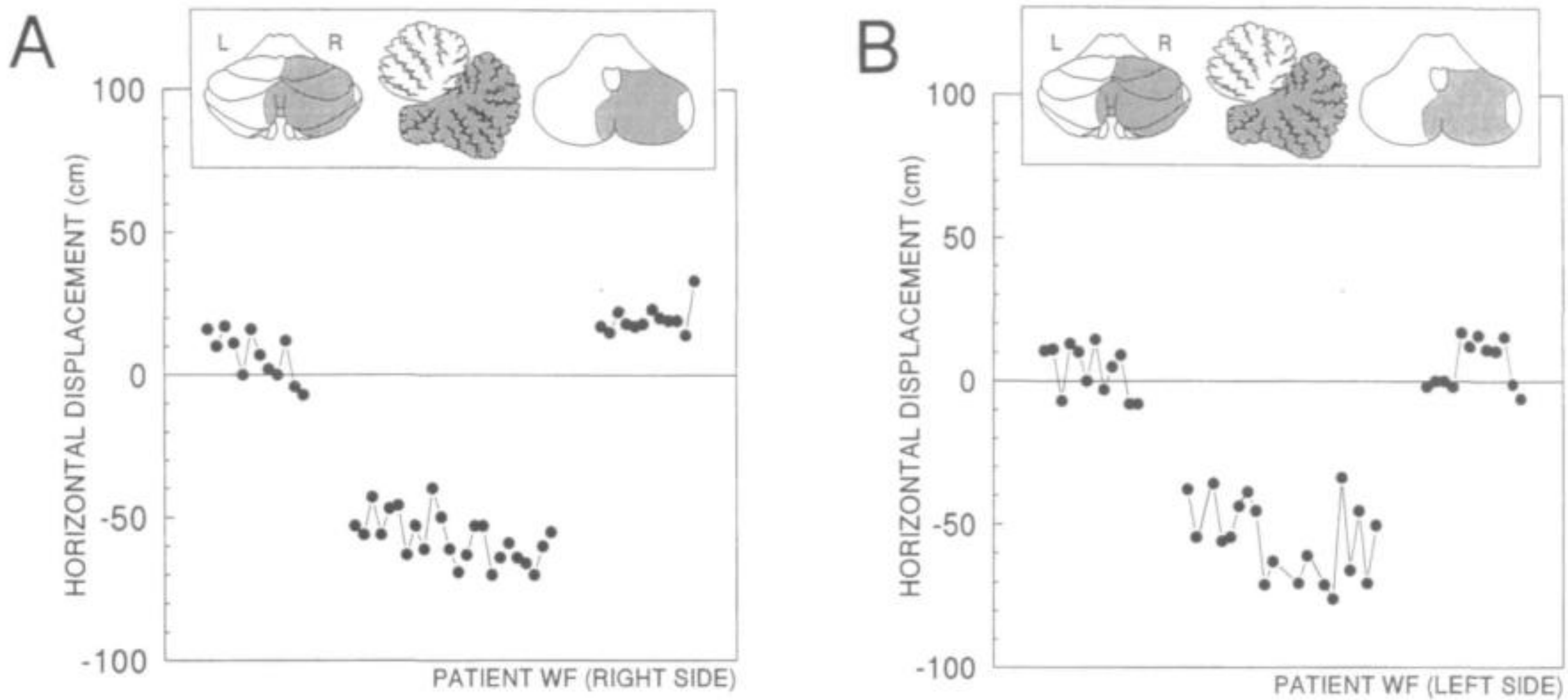
Rotate a cursor relative to movement (no vision of hand or arm)

Normal learning in prism adaptation



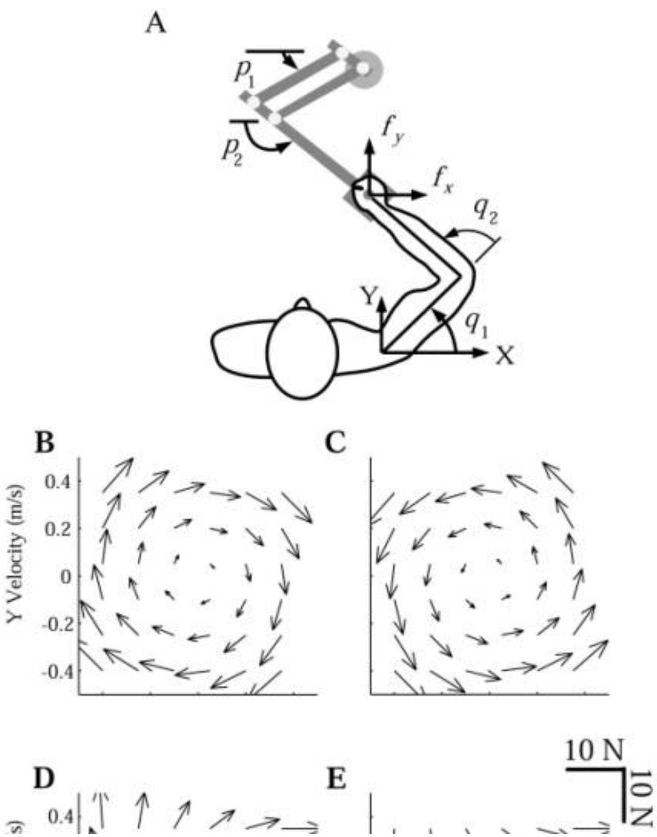
Martin *Brain* 1996 doi:10.1093/brain/119.4.1183

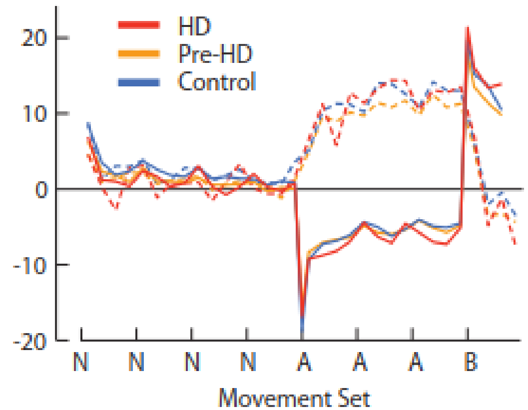
Impaired cerebellar learning

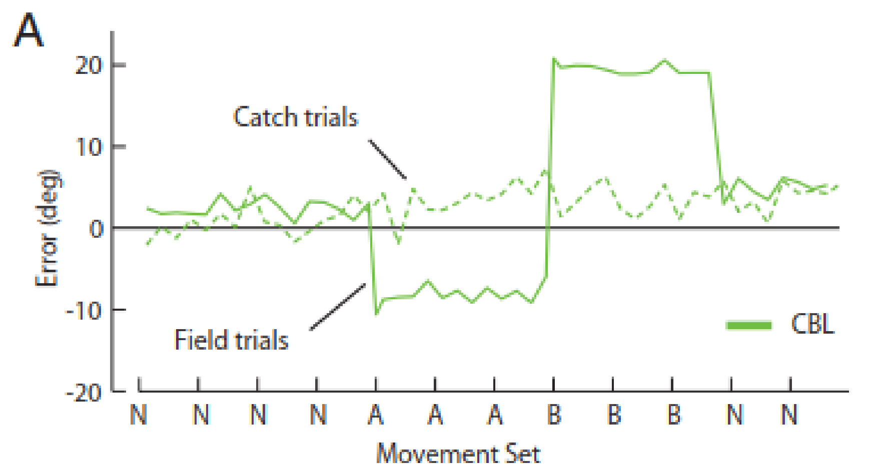


Martin *Brain* 1996 doi:10.1093/brain/119.4.1183

Force field adaptation

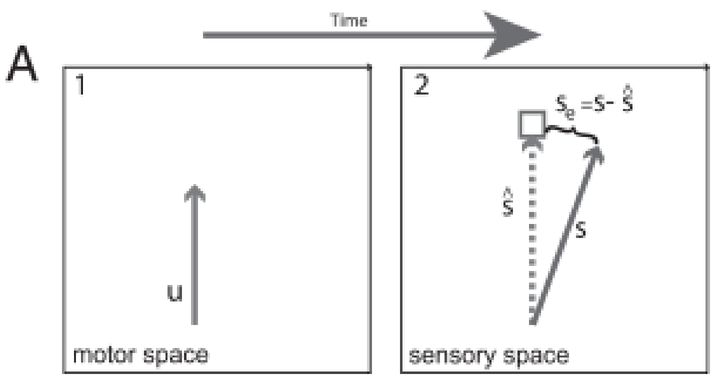


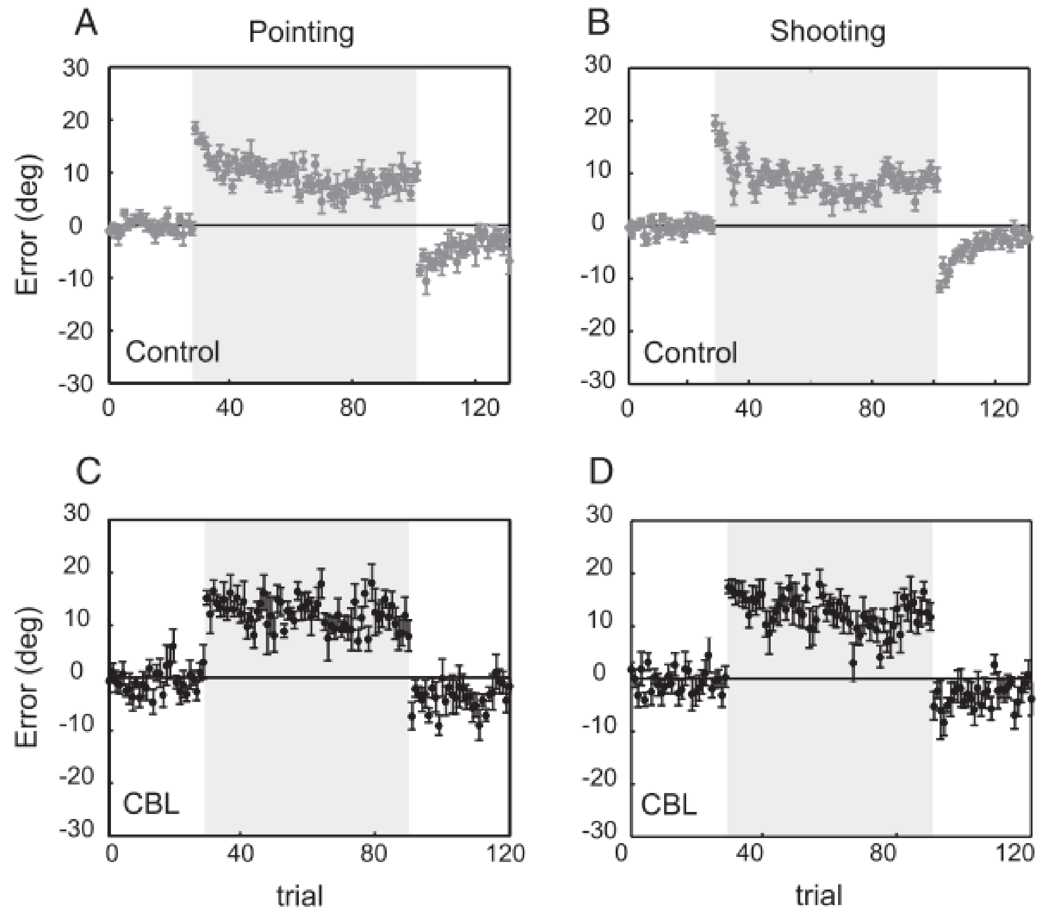




Smith *J Neurophysiol* 2004 doi:10.1152/jn.00943.2004

Visuomotor adaptation





Tseng *J Neurophysiol* 2007 doi:10.1152/jn.00266.2007

Other cerebellar dependent forms of adaptation

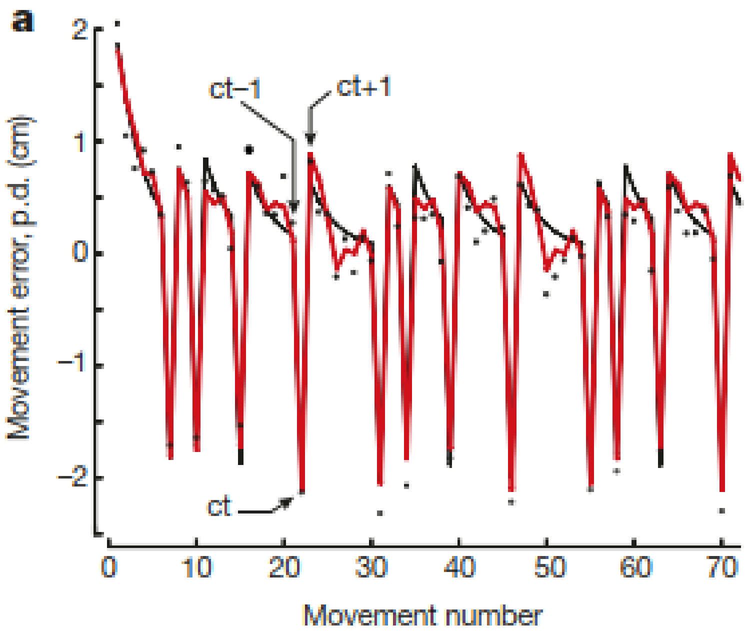
- Saccade adaptation
- Smooth pursuit adaptation
- Locomotor adaptation
- Split-belt treadmill adaptation

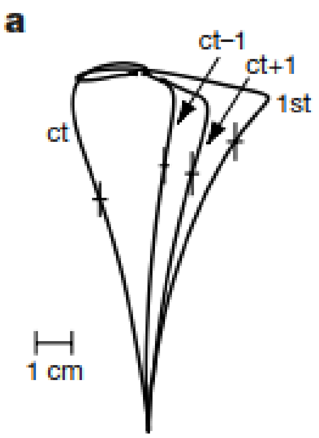
Split-belt treadmill

Part 4

- Modelling adaptation

Trial-by-trial adaptation





Thoroughman *Nature* 2000 doi:10.1038/35037588

Model of adaptation

$$\begin{cases} x_{i+1} = x_i - Be_i + \eta_i \\ e_i = x_i - p_i + \varepsilon_i \end{cases}$$

x_i: internal model of task

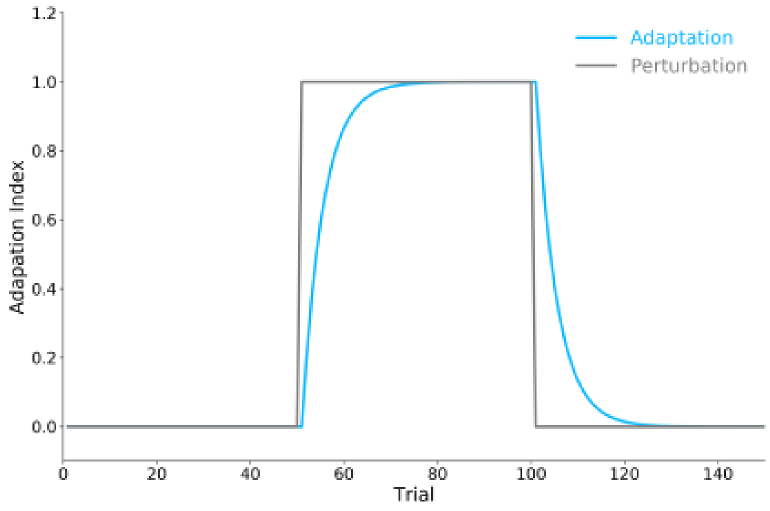
e_i: error in the movement

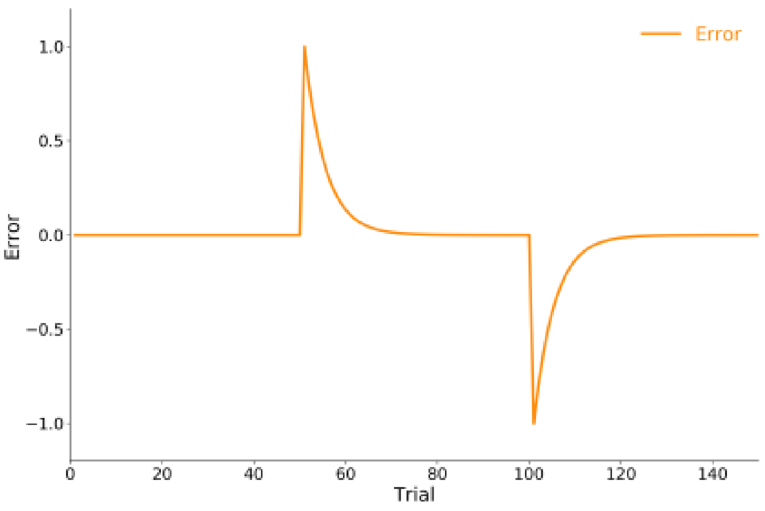
$\eta_i \sim N\left(0, \sigma_\eta^2\right)$: noise in the brain

$\varepsilon_i \sim N\left(0, \sigma_\varepsilon^2\right)$: noise in the output

B: learning rate

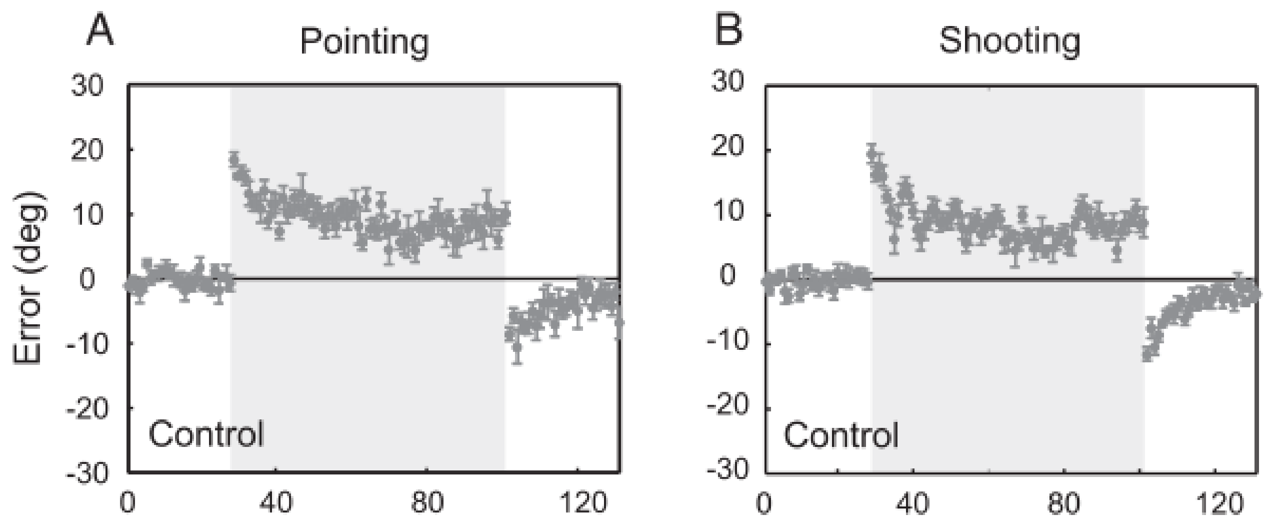
Simulating adaptation





$B = 0.2$

Model fails to explain incomplete learning



Tseng *J Neurophysiol* 2007 doi:10.1152/jn.00266.2007

Better model of adaptation

$$\begin{cases} x_{i+1} = Ax_i - Be_i + \eta_i \\ e_i = x_i - p_i + \varepsilon_i \end{cases}$$

x_i: internal model of task

e_i: error in the movement

$\eta_i \sim N\left(0, \sigma_\eta^2\right)$: noise in the brain

$\varepsilon_i \sim N\left(0, \sigma_\varepsilon^2\right)$: noise in the output

A: forgetting factor

B: learning rate

Steady state determined model

$$\begin{cases} x_{i+1} = Ax_i - Be_i + \eta_i \\ e_i = x_i - p_i + \varepsilon_i \end{cases}$$

In steady state:

$$\langle x_{i+1} \rangle = \langle Ax_i - Be_i + \eta_i \rangle = \langle x_i \rangle$$

$$A\langle x_i \rangle - B\langle e_i \rangle + \langle \eta_i \rangle = \langle x_i \rangle$$

$$A\langle x_i \rangle - B\langle e_i \rangle = \langle x_i \rangle$$

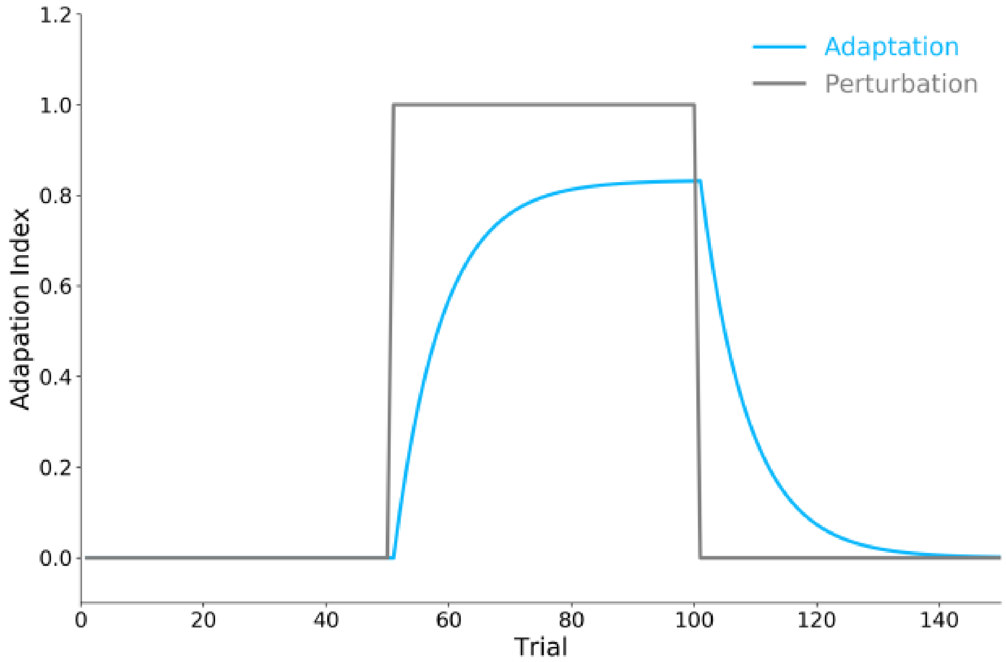
$$A\langle x_i \rangle - B\left(\langle x_i \rangle - p\right) = \langle x_i \rangle$$

$$\langle x_i \rangle = \frac{B}{1 - A + B} p$$

For instance, for $B = 0.2$ and $A = 0.98$ we get:

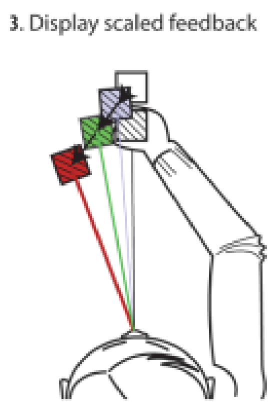
$$\langle x_i \rangle = \frac{0.1}{0.02 + 0.1} p = 0.83 p$$

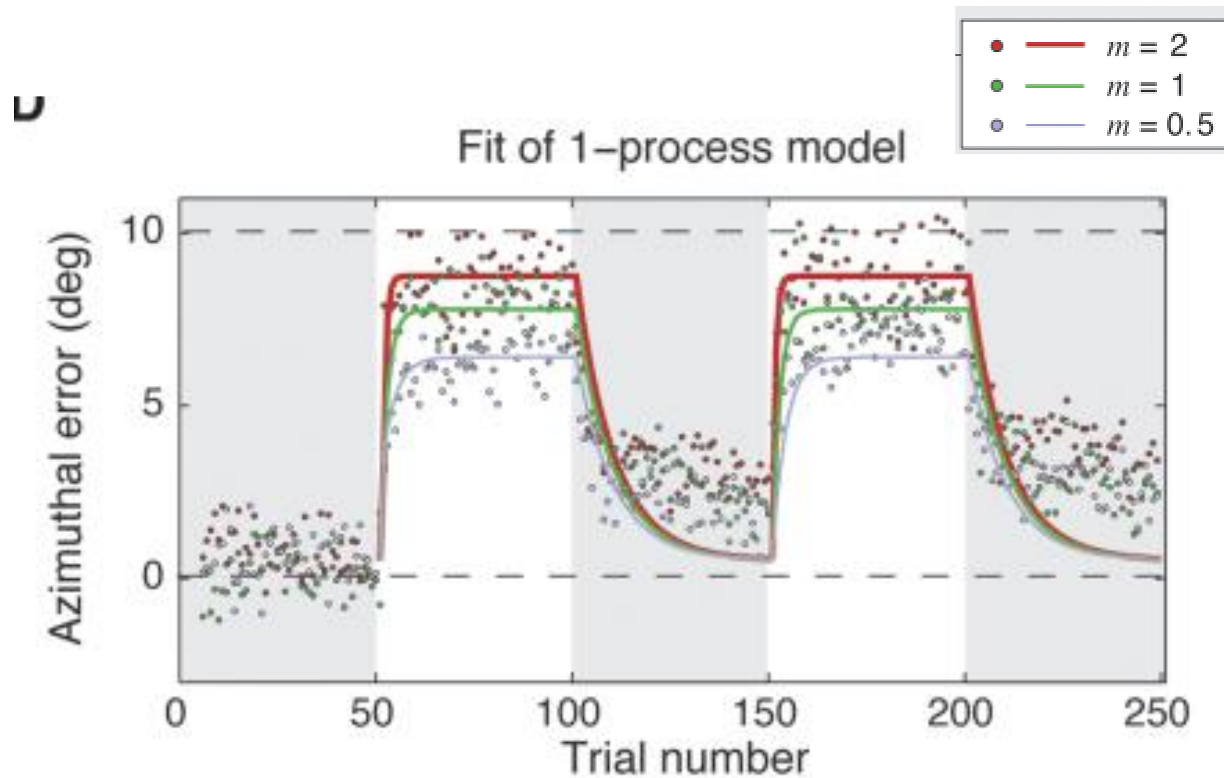
Model with forgetting



$$A = 0.98, B = 0.1$$

Increasing error magnitude effects learning but not forgetting



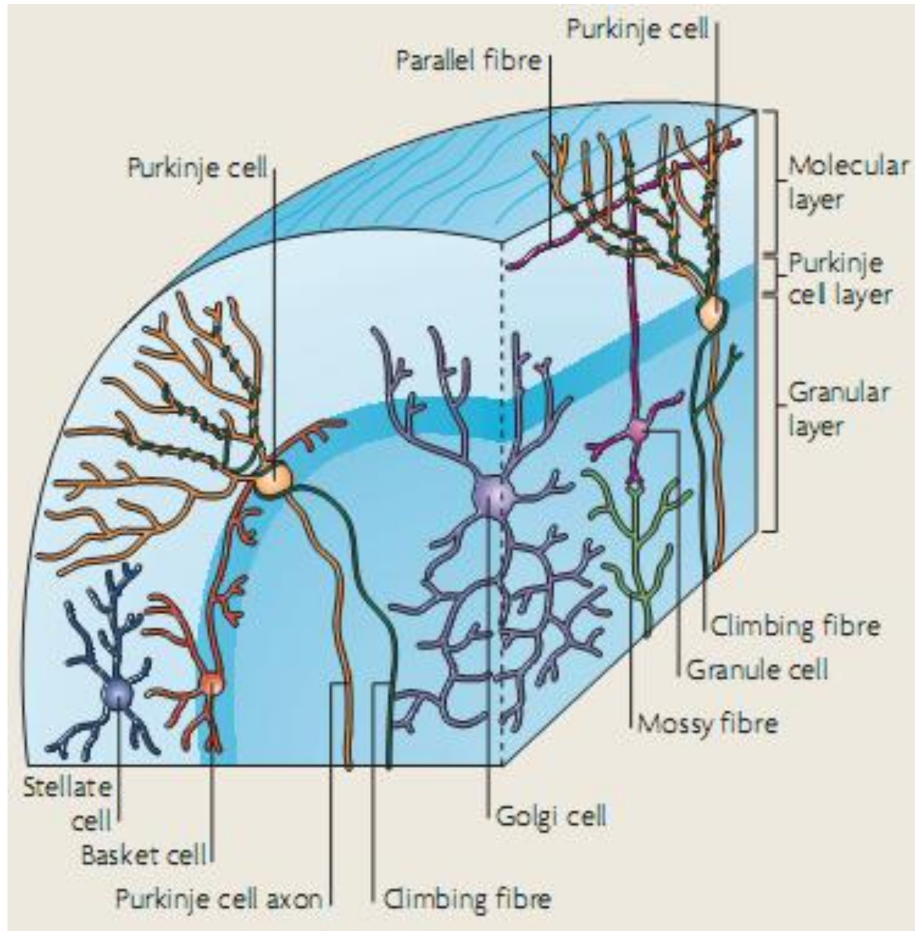


van der Kooij *PLOS One* 2015 doi:10.1371/journal.pone.0117901

Part 5

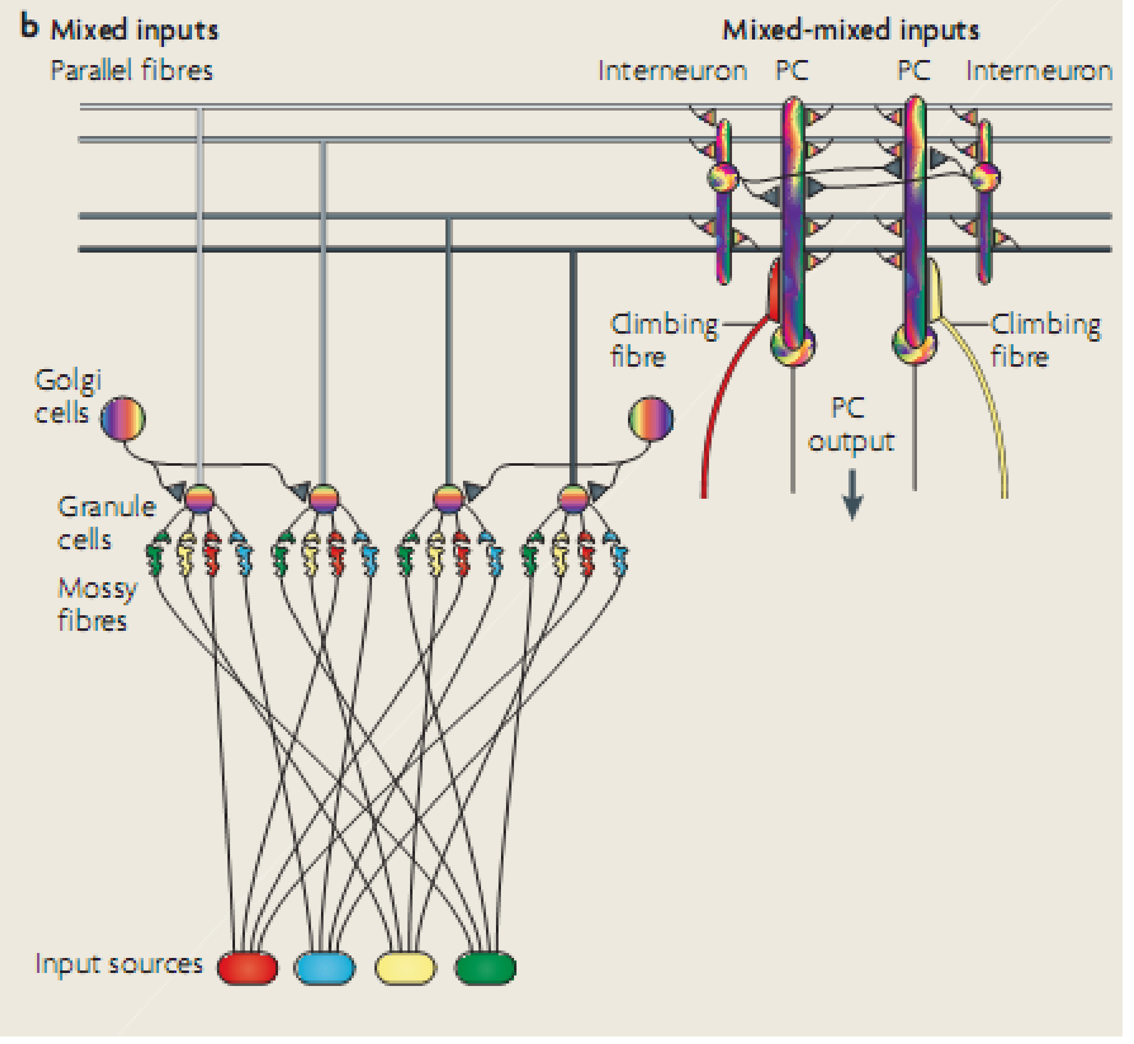
- The cerebellum and the model

Cerebellar anatomy



Dean *Nature Reviews Neuro* 2010

Schematic of input



Dean *Nature Reviews Neuro* 2010

The cerebellar model